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Mechanism of macrophage activation induced by polysaccharide from *Cordyceps militaris* culture broth

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Highlights

- Polysaccharides isolated from *C. militaris* (PLCM) have potent immunostimulating activity.
- PLCM was shown to increase NO, ROS, TNF-α, and phagocytic uptake capacity in RAW264.7 macrophages.
- The levels of nuclear NF-κB increased in LPS- or PLCM-treated RAW264.7 macrophages.
- TLR-4, Dectin-1, CR3 and TLR-2 are all involved in PLCM-induced TNF-α secretion.

Abstract

Mushroom-derived polysaccharides have been shown to stimulate immune responses. Our previous report showed that the novel polysaccharide PLCM isolated from the culture broth of *Cordyceps militaris* could induce nitric oxide production in the murine macrophage-like cell line RAW264.7. In this study, we show that PLCM enhances immunostimulatory activities such as the release of toxic molecules (nitric oxide and reactive oxygen species), secretion of the cytokine tumor necrosis factor (TNF)-α, and phagocytic uptake in RAW264.7 macrophages. In addition, all the specific inhibitors against the mitogen-activated protein kinase (MAPK) and nuclear factor kappa B (NF-κB) (SN50, BAY11-7082, PD98059, SP600125 and SB203580) markedly suppressed the nitric oxide production and phagocytic uptake induced by PLCM. Moreover, antibodies specific to the extracellular domain of Toll-like receptor-2, Toll-like receptor-4 or the macrophage receptor Dectin-1 significantly attenuated PLCM-induced secretion of TNF-α. Our results indicate that the *C. militaris* polysaccharide activates macrophages through the MAPKs and NF-κB signaling pathways *via* Toll-like receptor 2, Toll-like receptor 4, and Dectin-1.

Introduction

Macrophages are cells produced by the differentiation of monocytes in tissues in response to inflammatory stimuli or injury. They are specialized cells that neutralize foreign substances, infectious microbes and cancer cells through phagocytosis (Dentan, Lesnik, Chapman & Ninio, 1996). They function in non-specific defense (innate immunity) as well as help initiate specific defense mechanisms (adaptive immunity). Macrophages defend against external pathogens by releasing cytotoxic and inflammatory molecules such as nitric oxide (NO), reactive oxygen species (ROS) and secreting pro-inflammatory cytokines including tumor necrosis factor (TNF)-α and interleukin (IL)-1 (Grimes et al., 2005, Yoon et al., 2009). These defensive mechanisms are activated when various molecules that act as stimuli bind to pattern recognition receptors (PRRs) such as Toll-like receptors (TLRs), Dectin-1 and complement receptor type 3 (CR3 or CD11b/CD18) on the surface of macrophages and trigger several different signaling pathways including tyrosine kinases, protein kinase C, phosphoinositide-3-kinase (PI3K)/Akt, mitogen-activated protein kinases (MAPKs), and nuclear factor kappa B (NF-κB) (Biswas and Sodhi, 2002, Kuan et al., 2012, Wu et al., 2009).

Mushrooms have been shown to have several therapeutic effects (Bobek et al., 1991, Joseph et al., 2012, Selegean et al., 2009). Most, if not all, mushrooms have biologically active polysaccharides in the fruiting body, culture broth, and cultured mycelia. Many studies have demonstrated that polysaccharides from mushrooms are able to boost the host immune system by stimulating natural killer cells, T-cells, B-cells, and macrophages (Hou et al., 2009, Kim et al., 2008). In a previous study, we identified a novel exo-polysaccharide in the liquid culture broth of *Cordyceps militaris* (PLCM) (Lee et al., 2010b). We also showed that PLCM could induce NO production in the murine macrophage-like cell line RAW264.7.

In the present study, we examined whether PLCM could induce or enhance other immunostimulatory activities in macrophages and attempted to identify the mechanisms underlying macrophage activation by PLCM.

Section snippets

Materials

Fetal bovine serum (FBS), penicillin G, streptomycin, and RPMI 1640 were purchased from GIBCO (Grand Island, NY, USA). 3-(4,5-Dimethylthiazol-2-yl)-2,5-diphenyltetrazolium bromide (MTT), isopropyl alcohol, lipopolysaccharide (LPS, *Escherichia coli* 0111:B4), Polymyxin B(PMB), and fluorescein isothiocyanate (FITC)-labeled dextran were obtained from Sigma Chemical Co. (St. Louis, MO, USA). Dichlorodihydrofluorescein diacetate (H_2DCFDA), fluorescein-labeled *E. coli* (FITC-*E. coli*) bioparticles and …

PLCM induces the production of NO and ROS in macrophages

To investigate the effect of PLCM on the viability of RAW.264.7 macrophages, cells were treated with different concentrations of PLCM (10–1000 μg/ml) for 24 h, and then cell viability was determined using the MTT assay. PLCM at concentrations of 10–200 μg/ml did not affect cell viability significantly (Fig. 1A). To investigate whether PLCM was capable of inducing NO production from macrophages, RAW264.7 cells were treated with PLCM (10–1000 μg/ml) or LPS (2.5 μg/ml, positive control) for 24 h. PLCM …

Discussion

Immunostimulation is regarded as an important strategy for improving the body's defense mechanism in elderly people as well as in cancer patients (Rihova, Kovar, Kovar & Hovorka, 2009). Biological response modifiers isolated from mushrooms are being used increasingly to treat a wide variety of clinical conditions, albeit with relatively little knowledge of their modes of action (Yoshida et al., 2012). In particular, some of the major active substances in mushrooms are polysaccharides. …

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